清华大学气候变化与可持续发展研究院
Institute of Climate Change and Sustainable Development,Tsinghua University

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Assessment of Iron and Steel Technologies from A Global Perspective

Tsinghua University & Saudi Aramco

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清华大学气候变化与可持续发展研究院
Institute of Climate Change and Sustainable Development,Tsinghua University

Presenter: Yunchuan Liu, PhD student

January 26, 2026

Project Timeline

清华大学气候变化与可持续发展研究院
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	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	June	July	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	June	July
Research Proposal & preparations																								
Overview of global steel trade & technology assessment																								
Data requirement & database investigation (Base on case India)																								
Model upgrade (Base on case India)																								
Case study: India																								
Case study: USA & Saudi Arabia																								
Case study: Australia & Japan																								
Case study: Germany & Russia																								
Case study: China-Australia steel trade route																								
Final report																								

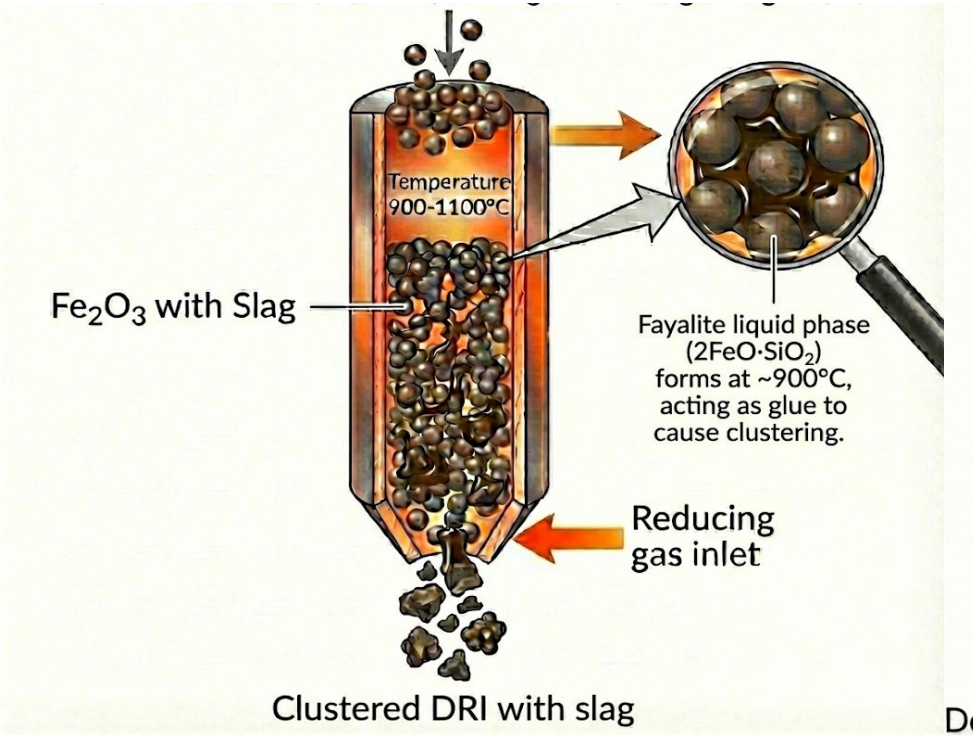
Research Update

- Supplementary information: Why can coal-based DRI utilize low-grade iron ore?
- Model update and analysis integrating plant-level data
- Outlook: Macroeconomic model of India

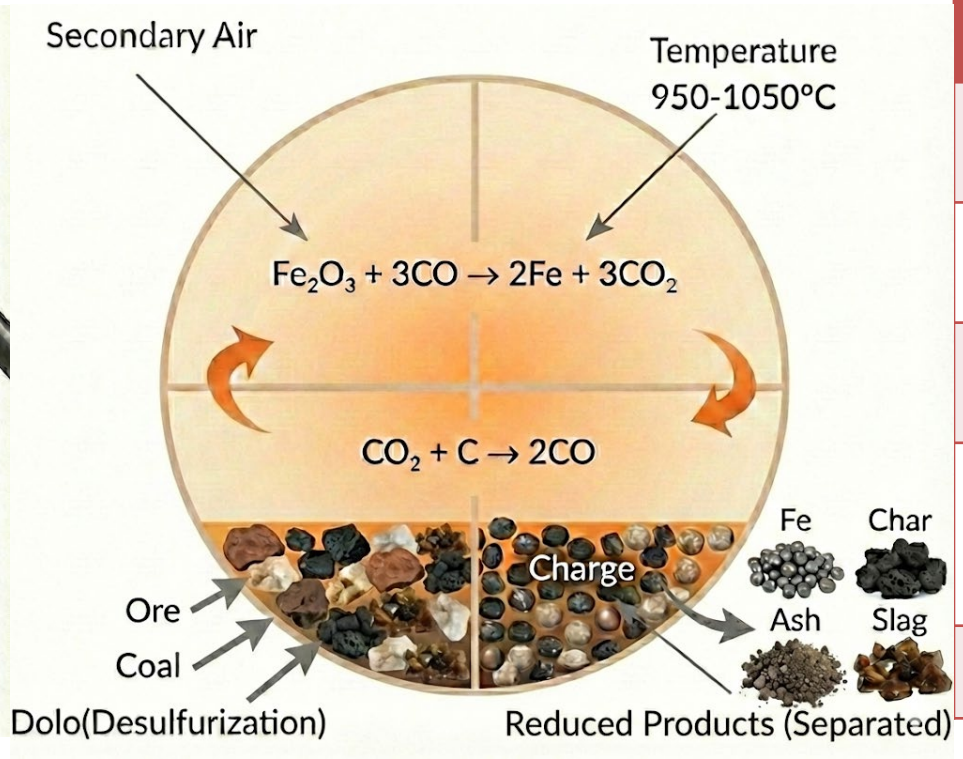
Why can coal-based DRI utilize low-grade iron ore?

- H2-DRI and NG-DRI are suitable for **shaft furnace**; Coal-DRI is suitable for **rotary kiln**.
- When the reaction temperature exceeds 900°C, a small amount of **liquid material** forms between the iron ore and slag, causing the small DRI particles to agglomerate with the slag into pellets.
- Because the solids in the rotary kiln are constantly tumbling, the presence of liquid can be tolerated without agglomeration, allowing for direct magnetic separation afterward.

Shaft furnace for H2-DRI/NG-DRI



Rotary kiln for Coal-DRI



Tech route	Ore Fe%	Ore Slag%
H2-DRI-EAF	**≥67-68%**	**<2-3%**
Gas-DRI-EAF	≥67%	<2-3%
Coal-DRI-EAF	≥64-65%	<4-6%
H2-DRI-ESF-BOF/EAF	**≥62-65%**	**/**
BF-BOF	≥58%	/

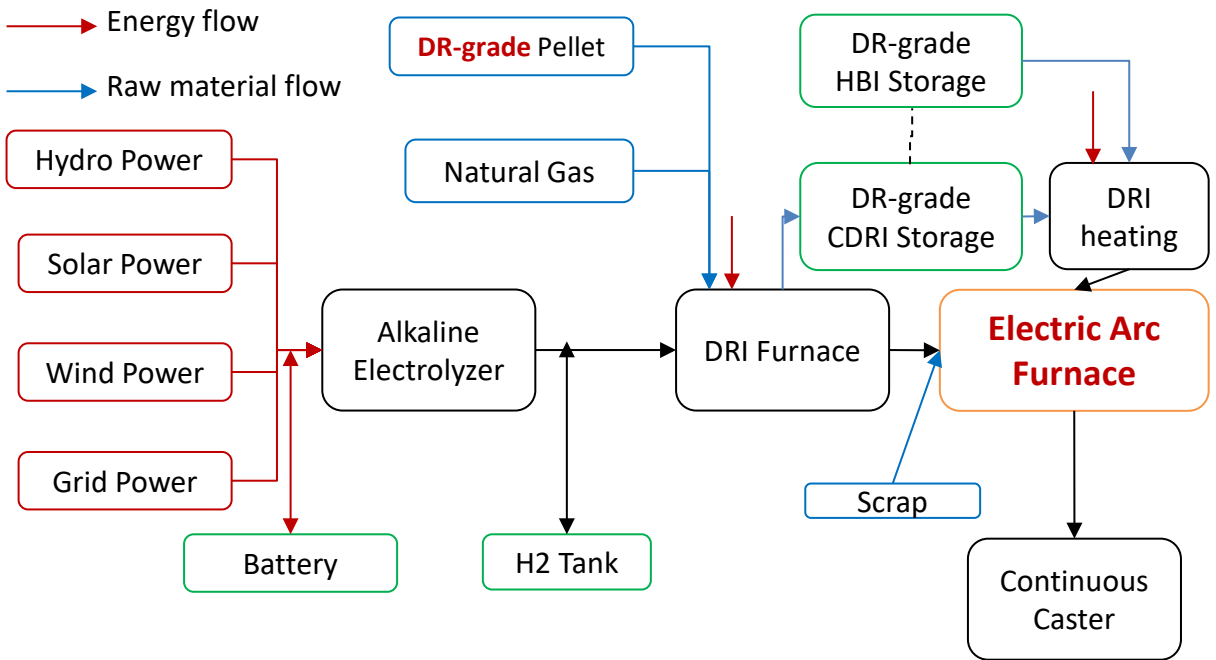
Source: Thyssenkrupp, BHP Billiton

Methods: updated model

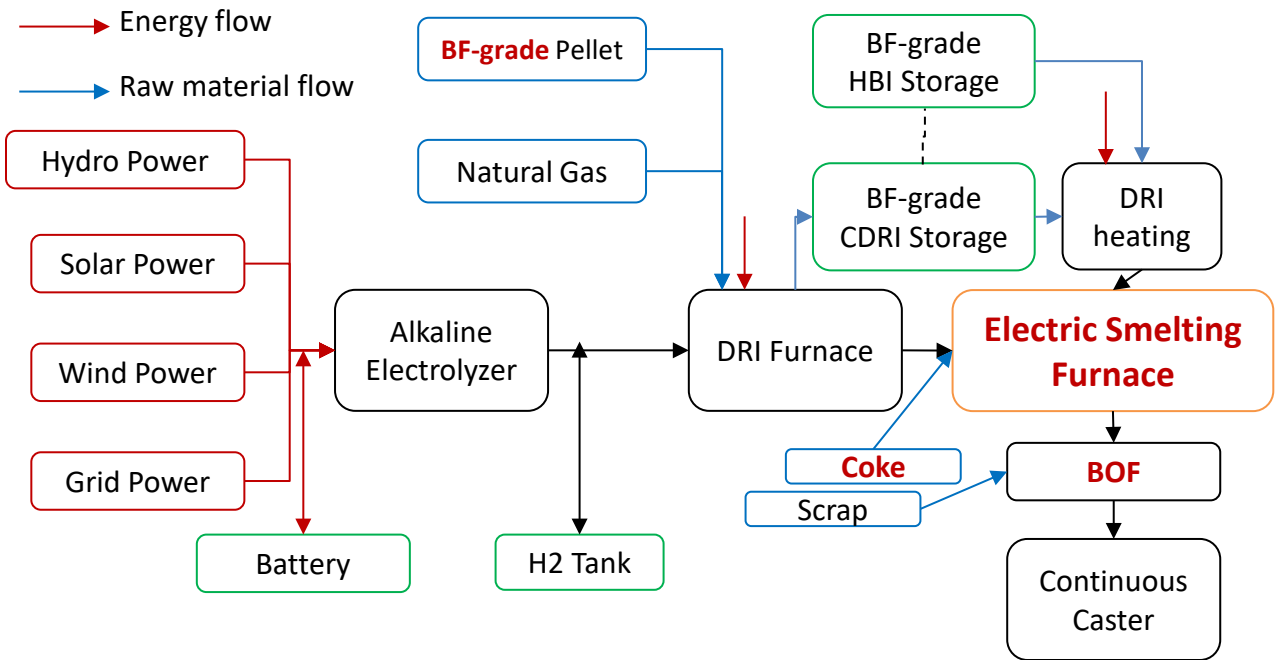
- Now we model the H2-DRI-ESF-BOF process specifically…

Main differences…	H2-DRI-EAF	H2-DRI-ESF-BOF
Feedstock	DR pellet	BF pellet & coke(**reductant**)
CAPEX	EAF	ESF+BOF(**Greenfield only**)
Energy consumption	EAF	ESF
Scrap is used for…	additional flexibility	**Balancing BOF temperature**
DRI supplement is…	flexible	**always same**

Traditional H2-DRI-EAF route

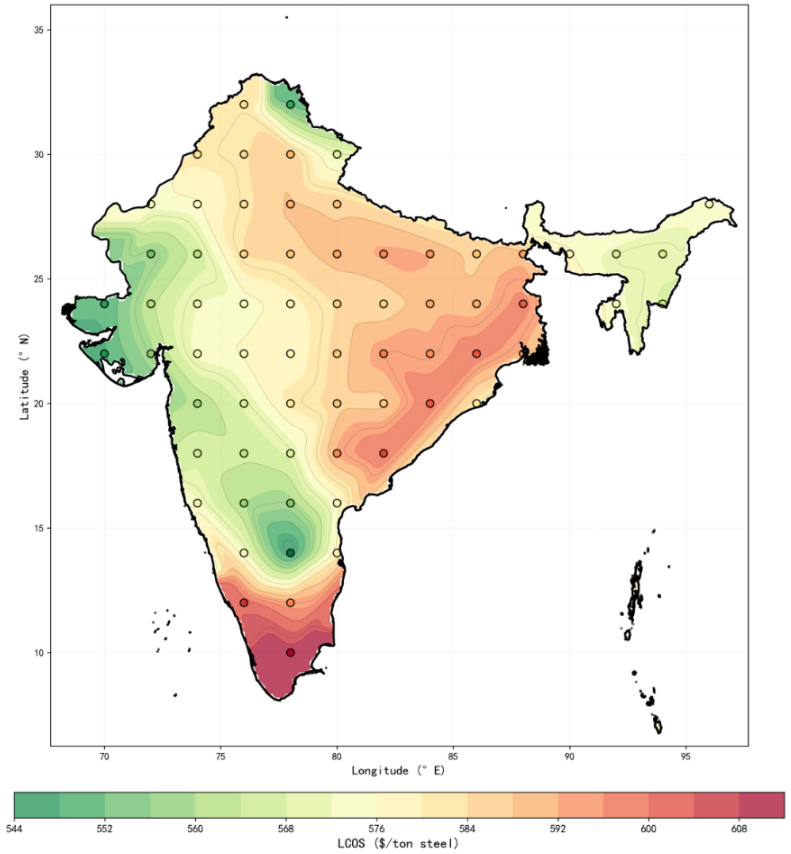


H2-DRI-ESF-BOF route

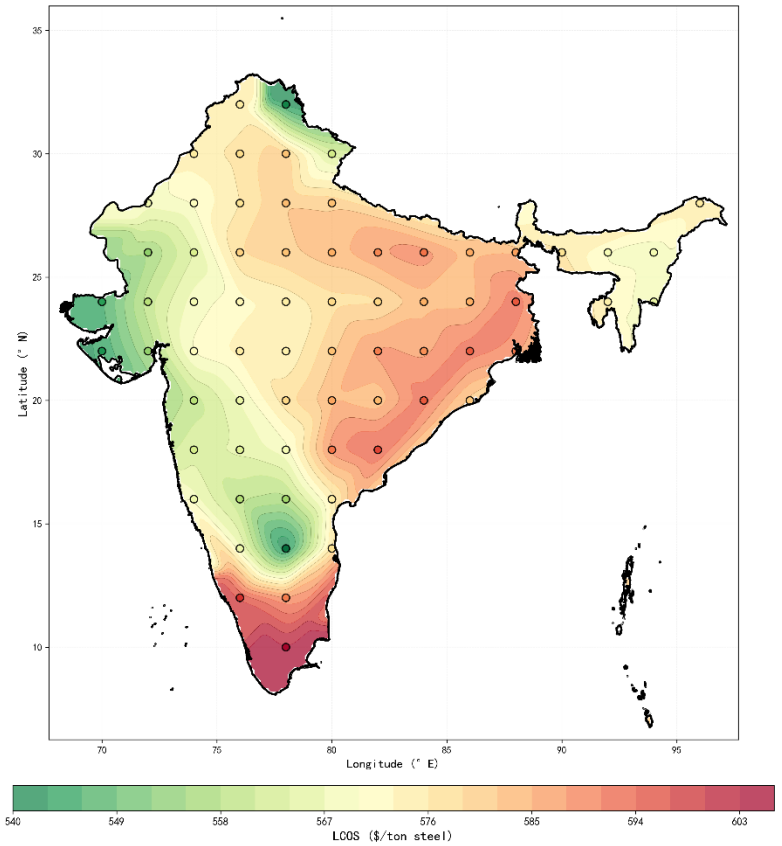


Updated results

- LCOS decreased for about **5-7 USD (almost unchanged,** *because we use the average value of domestic and import ore 108$/t*)
- Since the power output curve and load level did not change, the distribution of the results did not change so significantly.



LCOS heatmap of the H_2-DRI-EAF route in major regions of India (USD/tcs)
Using DR-pellet 168USD/t



LCOS heatmap of the H_2-DRI-ESF-EAF/BOF route in major regions of India (USD/tcs)
Using BF-pellet 108USD/t

India's iron ore supply prices

Type	price	Source
import DR pellet	$158-168	Estimates based on the Platts pellet premium
import BF pellet	$112-118	CFR Kandla / Mundra
domestic BF pellet	$88-98	Odisha Index

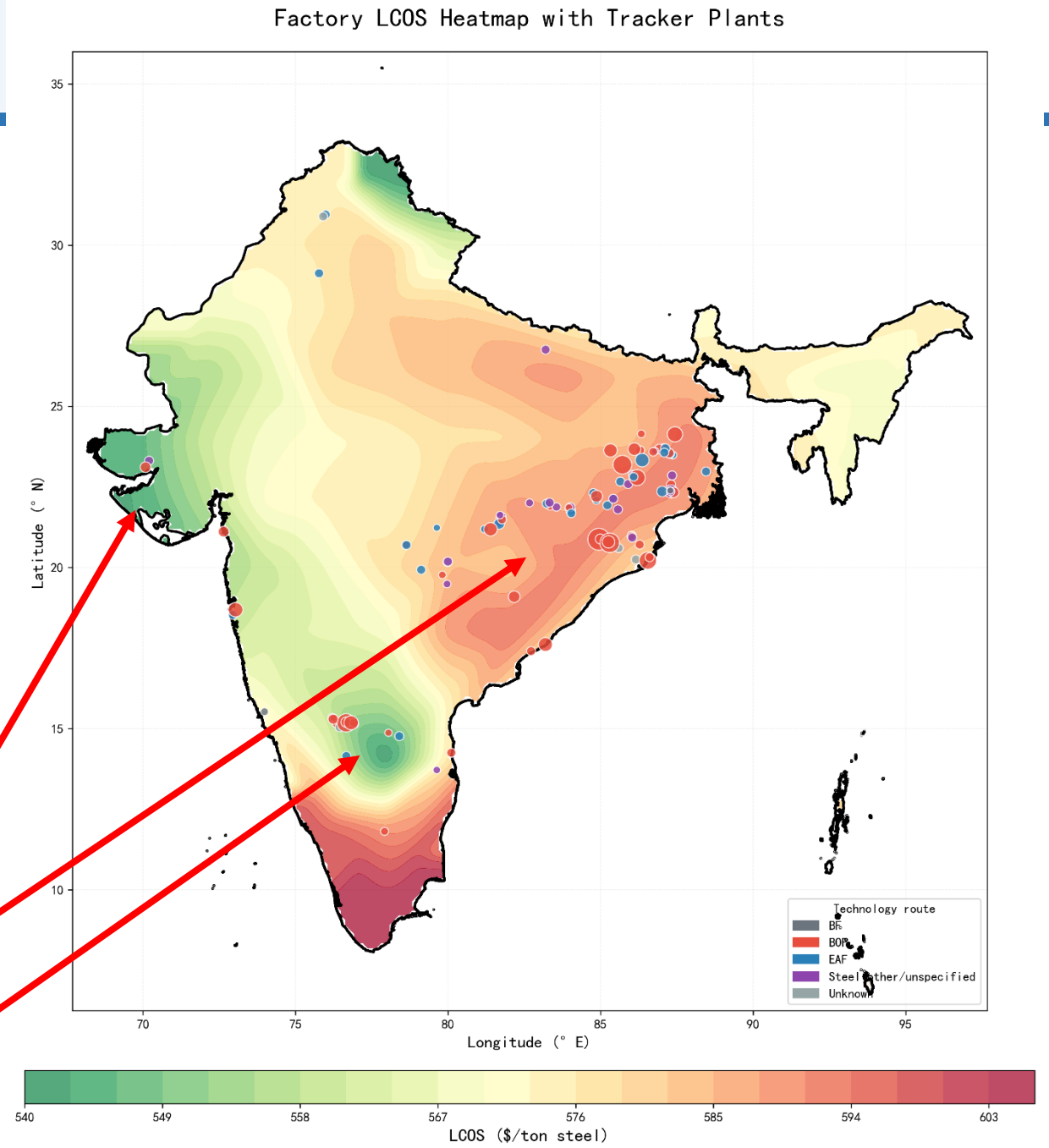
Analysis integrating plant-level data..

- India has basic conditions for **brownfield conversion** to H2-DRI-ESF-BOF.
- The LCOS of H2-DRI-ESF-BOF in the **eastern steel-producing region of India(Odisha, etc.)** is relatively high.
- The LCOS of H2-DRI-ESF-BOF in the steel-producing regions of **West coast of India(Gujarat)** and **south-central India** is relatively lower.
- **Brownfield conversion** in **west coast or south-central India** have the potential to be pioneers.

A. West coast of India(Gujarat, etc.)

B. Eastern India(Odisha, West Bangal, etc.)

C. South-central India(Andhra Pradesh, Karnataka, etc.)

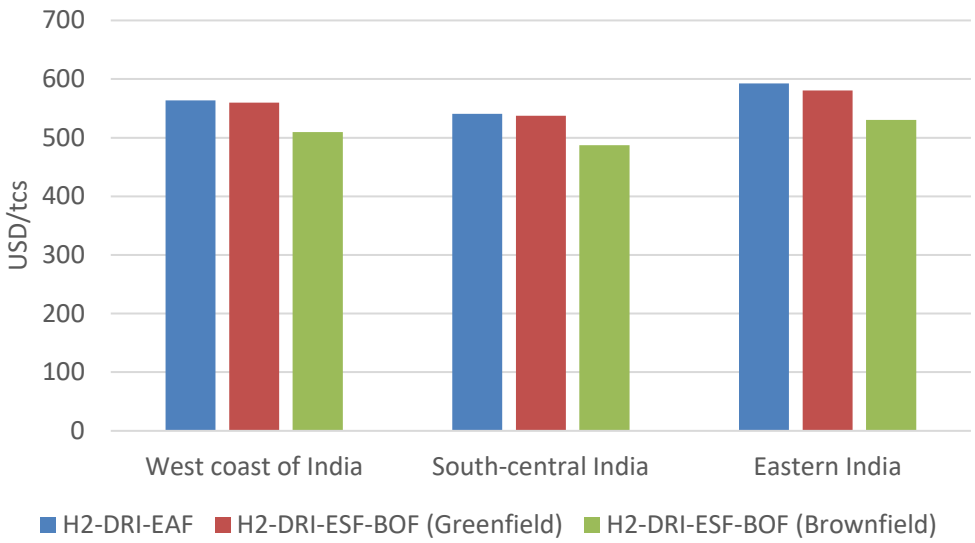


LCOS heatmap of **H_2-DRI-ESF-BOF** with **plant locations (>0.5Mt/y)**

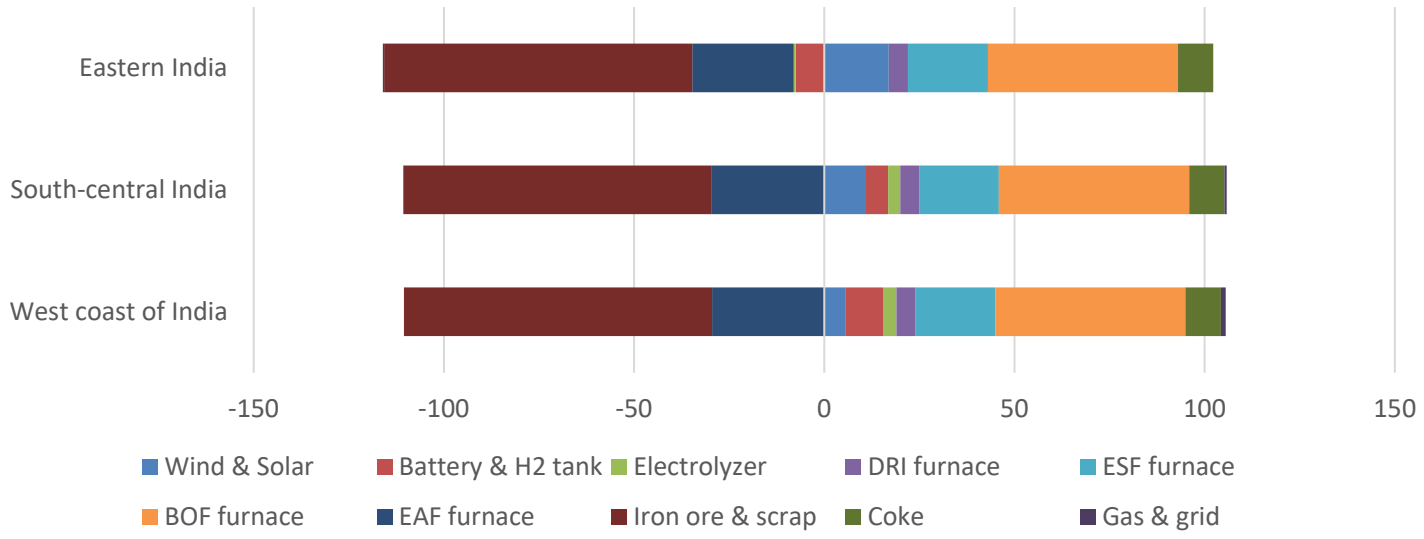
Cost structure analysis of typical locations

- For **new projects** or **greenfield developments**, the economics of building H2-DRI-EAF or H2-DRI-ESF-BOF are similar.
- For **existing BOF plants**, choosing H2-DRI-ESF-BOF brownfield conversion appears to be more cost-effective (saving 10% on LCOS).
- Compared to H2-DRI-EAF, the cost reduction in **iron ore & scrap** is crucial for H2-DRI-ESF-BOF.
- **Greenfield/brownfield development** and **price of iron ore & scrap** are key factors.

LCOS of interested green steel production technologies



Comparison of H2-DRI-EAF and **Greenfield** H2-DRI-ESF-BOF (USD/t)



Outlook: Macroeconomic model of India

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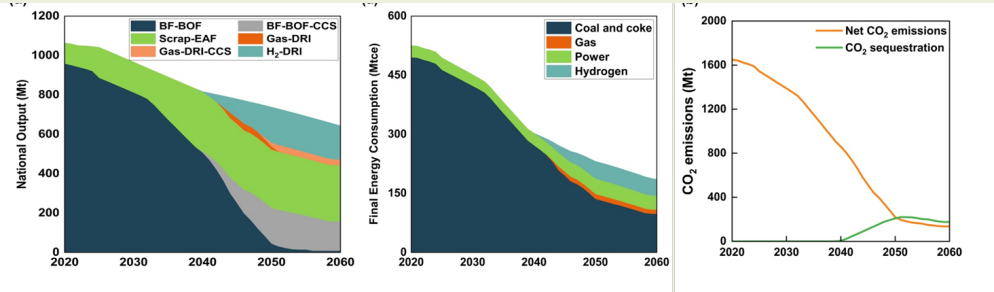
subject: India (nationwide)

Scenarios

Decarbonization pathway

- Capacity & Output
- Technology mix
- Energy & Emission
- Investment



Technologies

- BF-BOF
- Scrap-EAF
- Coal-DRI-EAF/IF
- H2-DRI-EAF
- H2-DRI-ESF-BOF
- …

Thank You For Your Attention!